Proteomics Code Challenges Week 2 | Step 1

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Despite many attempts, researchers have still not devised a scoring function that reliably assigns the highest score to the biologically correct peptide, i.e., the peptide that generated the spectrum. Fortunately, although the correct peptide often does not achieve the highest score among *all* peptides, it typically does score highest among all peptides *limited* to the species's proteome. As a result, we can transition from peptide sequencing to peptide identification by limiting our search to peptides present in the proteome, which we concatenate into a single amino acid string *Proteome*.

Peptide Identification Problem: *Find a peptide from a proteome with maximum score against a spectrum.*

Input: A spectral vector *Spectrum* and an amino acid string *Proteome*.

Output: An amino acid string *Peptide* that maximizes *Score*(*Peptide*, *Spectrum*) among all substrings of *Proteome*.

CODE CHALLENGE: Solve the Peptide Identification Problem.

Given: A space-delimited spectral vector *Spectrum'* and an amino acid string *Proteome*.

Return: A substring of *Proteome* with maximum score against *Spectrum'*.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/peptide_identification.txt)

Sample Input:

```
0 0 0 4 -2 -3 -1 -7 6 5 3 2 1 9 3 -8 0 3 1 2 1 8
XZZXZXXXZXZZXZXXZ
```

Sample Output:

```
ZXZXX
```

Time Limit: `5 mins`

To solve this problem please visit stepic.org/lesson/-11866/step/2

Like peptide sequencing algorithms, peptide identification algorithms may return an erroneous peptide, particularly if the score of the highest-scoring peptide found in the proteome is much lower than the score of the highest-scoring peptide over all peptides. For this reason, biologists usually establish a score threshold and only pay attention to a solution of the Peptide Identification Problem if its score is at least equal to the threshold.

Given a set of spectral vectors *SpectralVectors*, an amino acid string *Proteome*, and a score threshold *threshold*, we will solve the Peptide Identification Problem for each vector *Spectrum'* in *SpectralVectors* and identify a peptide *Peptide* having maximum score for this spectral vector over all peptides in Proteome (ties are broken arbitrarily). If *Score*(*Peptide*, *Spectrum*) is greater than or equal to *threshold*, then we conclude that *Peptide* is present in the sample and call the pair (*Peptide*, *Spectrum'*) a **Peptide-Spectrum Match (PSM)**. The resulting collection of PSMs for SpectralVectors is denoted $\text{PSM}_{threshold}$(*Proteome*, *SpectralVectors*).

PSM Search Problem: *Identify all Peptide-Spectrum Matches scoring above a threshold for a set of spectra and a proteome.*

Input: A set of spectral vectors *SpectralVectors*, an amino acid string *Proteome*, and an integer *threshold*.

Output: The set $\text{PSM}_{threshold}$(*Proteome*, *SpectralVectors*).

The following pseudocode solves the PSM Search Problem using an algorithm that you just implemented to solve the Peptide identification Problem, which we call **PeptideIdentification**.

PSMSearch(*SpectralVectors*, *Proteome*, *threshold*)
  PSMSet ← an empty set
  for each vector *Spectrum'* in *SpectralVectors*
    Peptide ← **PeptideIdentification**(*Spectrum'*, *Proteome*)
    if *Score*(*Peptide*, *Spectrum*) ≥ *threshold*
      add the PSM (*Peptide*, *Spectrum*') to *PSMSet*
  return *PSMSet*

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CODE CHALLENGE: Implement **PSMSearch** to solve the Peptide Search Problem.

Given: A set of space-delimited spectral vectors *SpectralVectors*, an amino acid string *Proteome*, and an integer *threshold*.

Return: The set $\text{PSM}_{threshold}(\textit{Proteome}, \textit{SpectralVectors})$.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/psm_search.txt)

Sample Input:

```
-1 5 -4 5 3 -1 -4 5 -1 0 0 4 -1 0 1 4 4 4
-4 2 -2 -4 4 -5 -1 4 -1 2 5 -3 -1 3 2 -3
XXXZXZXXZXZXXXZXXZX
5
```

Sample Output:

```
XZXZ
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11866/step/5

Imagine that we have an algorithm generating the set of all peptides scoring at least *threshold* against a spectral vector *Spectrum'*. We will henceforth call this set of high-scoring peptides a **spectral dictionary**, denoted $Dictionary_{threshold}(Spectrum')$. For a PSM (*Peptide*, *Spectrum'*), we will use the term **PSM dictionary**, denoted *Dictionary*(*Peptide*, *Spectrum'*), to refer to the spectral dictionary $Dictionary_{Score(Peptide,\ Spectrum')}(Spectrum)$.

Say we have a spectral dictionary consisting of a single amino acid string *Peptide*, which we will attempt to match against a randomly generated string *DecoyProteome* of length *n*. Because *DecoyProteome* was randomly generated, the probability that *Peptide* matches the string beginning at a given position of *DecoyProteome* is $1/20^{|Peptide|}$. We call this expression the **probability** of *Peptide*.

If the spectral dictionary consists of more than one peptide, then we define the **probability** of the dictionary as

$$\sum_{\text{each peptide Peptide in Dictionary}} \frac{1}{20^{|Peptide|}}.$$

Probability of Spectral Dictionary Problem: *Find the probability of a spectral dictionary for a given spectrum and a score threshold.*

Input: A spectral vector *Spectrum'* and an integer *threshold*.

Output: The probability of $Dictionary_{threshold}(Spectrum')$.

We will first compute the number of peptides in a spectral dictionary, since this simpler problem will provide insights on how to compute the probability of a spectral dictionary.

Size of Spectral Dictionary Problem: *Find the size of a spectral dictionary for a given spectrum and score threshold.*
 Input: A spectral vector *Spectrum'* and an integer *threshold*.
 Output: The number of peptides in $Dictionary_{threshold}(Spectrum')$.

We will use dynamic programming to solve the Size of Spectral Dictionary Problem. Given a spectral vector *Spectrum'* = $(s_1, \ldots, s_m)$, we define its ***i*-prefix** (for *i* between 1 and *m*) as $Spectrum'_i = (s_1, \ldots, s_i)$ and introduce a variable *Size*(*i*, *t*) as the number of peptides *Peptide* of mass *i* such that $Score(Peptide, Spectrum'_i)$ is equal to *t*.

The key to establishing a recurrence relation for computing *Size*(*i*, *t*) is to realize that the set of peptides contributing to *Size*(*i*, *t*) can be split into 20 subsets depending on their final amino acid *a*. Each peptide ending in a specific amino acid *a* results in a shorter peptide with mass $i - |a|$ and score $t - s_i$ if we remove a from the peptide (here, |a| denotes the mass of a). Thus,

$$Size(i,t) = \sum_{\text{all amino acids } a} Size(i-|a|, t-s_i)$$

Since there is a single “empty” peptide of length zero, we initialize *Size*(0, 0) = 1. We also define *Size*(0, *t*) = 0 for all possible scores *t*, and set *Size*(*i*, *t*) = 0 for negative values of i. Using the above recurrence, we can compute the size of a spectral dictionary of *Spectrum'* = $(s_1, \ldots, s_m)$ as

$$|Dictionary_{threshold}(Spectrum)| = \sum_{t \geq threshold} Size(m,t).$$

CODE CHALLENGE: Solve the Size of Spectral Dictionary Problem.

Given: A spectral vector *Spectrum'*, an integer *threshold*, and an integer *max_score*.

Return: The size of the dictionary $Dictionary_{threshold}(Spectrum')$.

Note: Use the 20 amino acid alphabet as well as the provided *max_score* for the height of your table. Your answer should be the number of peptides whose score is at least *T* and at most *max_score*.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/size_spectral_dictionary.txt)

Sample Input:

```
4 -3 -2 3 3 -4 5 -3 -1 -1 3 4 1 3
1
8
```

Sample Output:

```
3
```

Time Limit: `5 mins`

To solve this problem please visit stepic.org/lesson/-11866/step/8

Note that the equation for the *probability* of a dictionary,

$$\Pr(Dictionary) = \sum_{\text{each peptide } Peptide \text{ in } Dictionary} \frac{1}{20^{|Peptide|}},$$

is similar to an equation for the *size* of a dictionary,

$$|Dictionary| = \sum_{\text{each peptide } Peptide \text{ in } Dictionary} 1.$$

This similarity suggests that we can derive a recurrence for the probability of a dictionary using arguments similar to those used to find the size of a dictionary.

Define Pr(i, t) as the sum of probabilities of all peptides with mass i for which *Score*(*Peptide*, $Spectrum'_i$) is equal to t. The set of peptides contributing to Pr(i, t) can be split into 20 subsets depending on their final amino acid. Each peptide *Peptide* ending in a specific amino acid a results in a shorter peptide $Peptide_a$ if we remove a; $Peptide_a$ has mass $i - |a|$ and score $t - s_i$. Since the probability of *Peptide* is 20 times smaller than the probability of $Peptide_a$, the contribution of *Peptide* to Pr(i, t) is 20 times smaller than contribution of $Peptide_a$ to Pr($i - |a|$, $t - s_i$). Therefore, Pr(i, t) can be computed as

$$\Pr(i, t) = \sum_{\text{all amino acids } a} \frac{1}{20} \cdot \Pr(i - |a|, t - s_i)$$

which differs from the recurrence for computing *Size*(i, t) only in the presence of the factor 1/20. We can now compute the probability of a spectral dictionary as

$$\Pr(Dictionary_{threshold}(Spectrum')) = \sum_{t \geq threshold} \Pr(m, t).$$

CODE CHALLENGE: Solve the Probability of Spectral Dictionary Problem.

Given: A spectral vector *Spectrum'*, an integer *threshold*, and an integer *max_score*.

Return: The probability of the dictionary $Dictionary_{threshold}(Spectrum')$.

Note: Use the provided *max_score* for the height of your table.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/probability_spectral_dictionary.txt)

Sample Input:

```
4 -3 -2 3 3 -4 5 -3 -1 -1 3 4 1 3
1
8
```

Sample Output:

```
0.375
```

Time Limit: `5 mins`

To solve this problem please visit stepic.org/lesson/-11866/step/11

In addition to searching for mutated peptides, we will also need to search for **post-translational modifications**, which alter amino acids after a protein has been translated from RNA. In fact, most proteins are modified after translation, and hundreds of types of modifications have been discovered. For example, the enzymatic activity of many proteins is regulated by the addition or removal of a phosphate group at a specific amino acid. This process, called **phosphorylation**, is reversible; **protein kinases** add phosphate groups, whereas **protein phosphatases** remove them.

A modification of mass δ applied to an amino acid results in adding δ to the mass of this amino acid. For example, $\delta = 80$ for phosphorylated amino acids (serine, threonine, and tyrosine), $\delta = 16$ for the modification of proline into hydroxyproline, and $\delta = -1$ for the modification of lysine into allysin. If δ is positive, then the resulting modified peptide has a peptide vector that differs from the original peptide vector *Peptide'* by inserting a block of δ zeroes before the i-th occurrence of 1 in *Peptide'*. In the more rare case that δ is negative, the modified peptide corresponds to deleting a block of $|\delta|$ zeroes from *Peptide'*.

We will use the term **block indel** to refer to the addition or removal of a block of consecutive zeroes from a binary vector. Thus, applying *k* modifications to an amino acid string *Peptide* corresponds to applying *k* block indels to its peptide vector *Peptide'*. We define $Variants_k$(*Peptide*) as the set of all modified variants of *Peptide* with up to *k* modifications.

Given a peptide *Peptide* and a spectral vector *Spectrum'*, our goal is to find a modified peptide from $Variants_k$(*Peptide*) with maximum score against *Spectrum'*.

Spectral Alignment Problem: *Given a peptide and a spectral vector, find a modified variant of this peptide that maximizes the peptide-spectrum score, among all variants of the peptide with up to k modifications.*

Input: An amino acid string *Peptide*, a spectral vector *Spectrum'*, and an integer *k*.
Output: A peptide of maximum score against *Spectrum'* among all peptides in $Variants_k$(*Peptide*).

CODE CHALLENGE: Solve the Spectral Alignment Problem.

Given: A peptide *Peptide*, a spectral vector *Spectrum'*, and an integer *k*.

Return: A peptide *Peptide'* related to *Peptide* by up to *k* modifications with maximal score against *Spectrum'* out of all possibilities.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/spectral_alignment.txt)

Sample Input:

```
XXZ
4 -3 -2 3 3 -4 5 -3 -1 -1 3 4 1 -1
2
```

Sample Output:

```
XX(-1)Z(+2)
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11866/step/14